SUPERVISOR PLANNING MEETING

CHRONO-Fair

Time-Resolved Counterfactual Fairness Monitoring for Clinical Machine Learning

Extends our prior VFR-Audit three-axis static auditing framework

Target venue: Q1 digital-health / health-informatics journal

Anonymous draft | in-progress research | not finalised

Where clinical ML stands today

Clinical ML supports triage, risk scoring, and length-of-stay prediction across many sites [1, 2].

Documented post-release subgroup failures across radiology, sepsis, kidney function, and care-allocation models [3, 4, 5, 6].

Regulatory expectations have shifted toward continuous lifecycle monitoring [7, 8] with reporting standards [9, 10, 11, 12].

Auditing toolkits in routine use return a thresholded pass-or-fail verdict on a fixed snapshot [13, 14, 15, 16].

That verdict is the object hospitals act on. The snapshot tells us nothing about its stability over time.

Our prior VFR-Audit study

static auditing framework

Three audit-reliability axes on real Texas-100X (925 128 records) [17]:

- Axis 1: resampling fluctuation of the fairness verdict.
- Axis 2: audit-size sensitivity of the fairness verdict.
- Axis 3: cross-hospital-site agreement of the fairness verdict.

Standard vs Fair monitored-model setup (cited from prior work):

Attribute	Std DI	Fair DI	Verdict
Race	0.654	**0.884**	fail → **pass**
Sex	0.762	**0.933**	fail → **pass**
Ethnicity	0.832	**0.962**	pass → **pass**
Age Group	0.291	**0.801**	fail → **pass**

Std Acc 0.8777 → Fair Acc 0.8248 (reduction of 5.28 pp). Predecessor, not a CHRONO-Fair contribution.

The remaining gap

◎ **Real Texas-100X anchor finding (925 128 records) [17].** The worst-group true-positive-rate verdict for race reads "fair" at **0.824**. The same verdict flips to "unfair" in **4 of 20** hospital-network resamples (Verdict Flip Rate **0.20**). A batch audit cannot tell when, or whether, that verdict reversal recurs as new patients arrive.

Gap 1, time axis.
Batch verdicts give no timestamp [13, 14, 15].

Gap 2, peeking penalty.
Repeated inspection inflates false-alarm rates without a sequential guarantee [18, 19, 20].

Gap 3, diagnostic interpretation.
Without triage, a team cannot tell whether to fix data or collect more samples [21, 22].

Gap 4, regression output.
LOS outputs lack a flip-rate or verdict-stability extension [23].

Closest related work, named and positioned

Tool / work	Mode	Output	Closest gap
Aequitas [13]	Batch	Per-attribute parity-ratio report	No streaming alarm
Fairlearn [14]	Batch	Group fairness + reduction mitigator	No verdict-flip monitoring
AIF360 [15]	Batch	20 metrics library	Snapshot only
Microsoft RAI [16]	Notebook	Fairlearn + interpret + CF	Developer review, not gover-nance
Davis et al. [2]	Quarterly	Subgroup trend, 11 yr VA	No per-cell sequential alarm
Prediction sensitivity [22]	Per batch	CF-flip rate scalar	No alarm rule
VFR-Audit (our prior) [17]	Static	Verdict Flip Rate scalar	No time axis

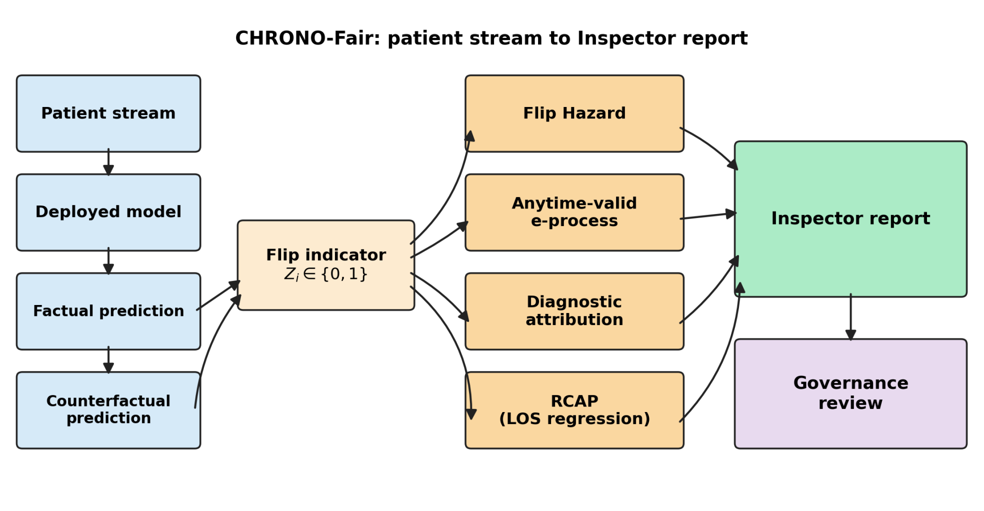
Anytime-valid machinery exists [18, 19, 20]. It has not been applied to counterfactual fairness verdicts at intersectional granularity.

Our planned extension, in one sentence

CHRONO-Fair contributes a post-release, time-resolved monitoring layer for thresholded fairness verdicts using a per-cell anytime-valid e-process on counterfactual flip indicators.

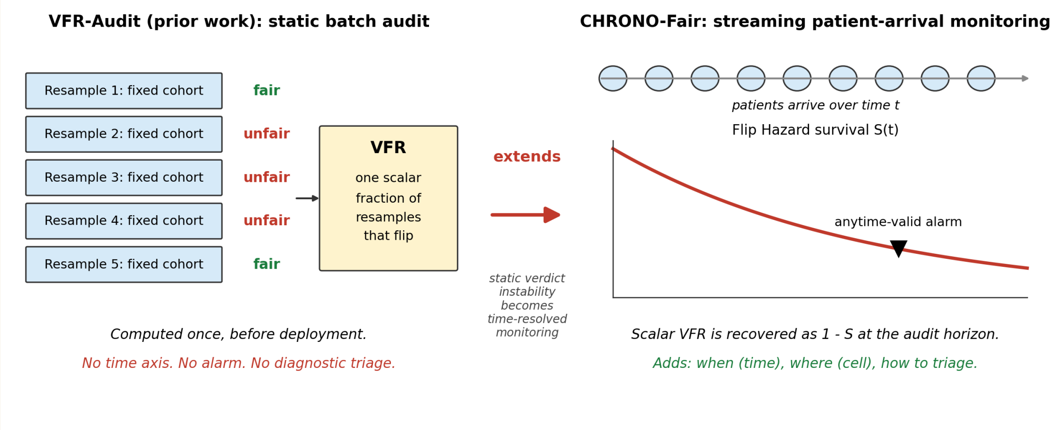
- Monitor, not mitigator
- Begins after Std $\rightarrow$ Fair
- Agnostic to mitigation method
- Research prototype, not medical device

Planned architecture



Modules: Flip Hazard [24]; per-cell shrunk-GROW e-process [18, 19]; aleatoric/epistemic decomposition [21]; RCAP Wasserstein-1 rank shift [23]

How CHRONO-Fair extends VFR-Audit



Scalar VFR is recovered as $1 - S$ at the audit horizon. CHRONO-Fair adds the time axis, the anytime-valid alarm rule, and the diagnostic triage signal.

What is new here

Five things are new, even though each statistical instrument is published elsewhere.

1. **New monitored object.** The *per-cell counterfactual fairness-verdict flip* as a time-indexed stream. Prior anytime-valid work [19, 20, 29] monitors aggregate risk. Prior fairness work [13, 14, 15, 22] reports static metrics. Verdict flips per cell over time is not a monitored quantity in either literature.
2. **Coordinated framework.** Flip Hazard ($\rightarrow$ *when*); per-cell e-process ($\rightarrow$ *alarm*); aleatoric/epistemic decomposition ($\rightarrow$ *triage*); RCAP ($\rightarrow$ *regression*). All four feed one indicator and one Inspector report.
3. **Shrunk-GROW bet λ_s + warm-up.** Finite-sample stabilisation of the GROW rule of [29] for the small-flip-rate regime typical of clinical fairness.
4. **RCAP statistic.** W_1 is textbook [23]; applying it to counterfactual rank-position distributions of an LOS-regression output is, to our knowledge, new.
5. **Anchor + controlled-replay evidence.** Real Texas-100X verdict instability at scale; then characterised under known drift.

Honest attribution: hazard form [24]; RMST; e-process wealth [29]; Ville's inequality [18]; aleatoric/epistemic split [21, 30]; Wasserstein-1 [23]; step-wise BH [26]. The contribution is the coordinated application, not the

How each planned module closes one identified gap

Gap 1, time axis	»	**Flip Hazard.** Per-arrival survival curve instead of a single scalar [24].
Gap 2, peeking penalty	»	**Per-cell anytime-valid e-process.** Type-I control at every inspection step [18, 19, 20].
Gap 3, diagnostic interpretation	»	**Aleatoric/epistemic decomposition.** Triage label (data, model, mixed) per alarmed cell [21].
Gap 4, regression output	»	**RCAP, Wasserstein-1 on ranks.** Counterfactual rank shift for LOS regression [23].

Method discipline: what we will NOT claim

- **Per-cell anytime-valid Type-I control:** YES, under Ville inequality [18, 19].
- **Time-uniform online FDR:** NO. Step-wise BH is applied at inspection time only [25, 26].
- **Causal root cause:** NO. Diagnostic attribution is aleatoric/epistemic triage [21].
- **SCM-strength counterfactual:** NO. Feature-shift only; may under-flag indirect effects [27, 28].
- **Prospective clinical deployment:** NO. Controlled replay and simulation only.

Anchor streaming result

controlled replay

Method	Detected	Mean delay (patients)
CHRONO-Fair (per-cell e-process)	**50 / 50**	**837 (median 828)**
ADWIN ($k = 1$)	21 / 50	3 878
ADWIN ($k = 25$)	21 / 50	3 927
Davis-style ADWIN on fairness gap	21 / 50	3 904
Periodic batch ($n = 500$)	24 / 50	499 (no peeking control)
Static VFR	0 / 50	–

10 000-patient stream, drift onset at 5 000, 50 Monte-Carlo runs. False-alarm rate at or below the nominal α for every tested α [19, 29].

Streaming evidence is controlled replay and simulation. Not prospective clinical validation.

Plan: how the gap will actually be closed

Phase 1 **Controlled-validation evidence.** Real Texas-100X batch + Texas-100X-calibrated streaming/replay + robustness stress tests.
Status: drafted, in revision for Q1 submission.

Phase 2 **Methodological strengthening.** Time-uniform online FDR [25, 26]; SCM counterfactual sensitivity [27, 28].
Status: planned for next revision.

Phase 3 **Prospective evaluation.** Site-partnered EHR-stream validation; governance-user evaluation.
Status: scoping, partner search.

Phase 4 **Artifact release.** Anonymised proof-of-concept prototype + configs + seeds + scripts.
Status: scheduled for camera-ready.

Asks for this meeting

1. Is the post-release monitoring framing the right contribution to lead with, vs a methods-only framing of the per-cell e-process?
2. Is the Std/Fair monitored-model table (cited from prior VFR-Audit study) acceptable as the predecessor without re-running it?
3. Which Q1 venue is best: Frontiers in Digital Health, npj Digital Medicine, JAMIA, JBI, or NEJM AI?
4. Should I add a second real cohort (MIMIC-IV / eICU) before submission, or submit single-dataset with a prospective-validation roadmap?
5. Defer artifact release to camera-ready, or supply a minimum-viable artifact preview at submission?
6. Timing: next-round submission in 6 to 8 weeks, or invest 2 to 3 months in Phase 2 strengthening first?

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